
Problem Set 6

Problem 4.5

How much heat is required when 10,000 kg of CaCO_3 is heated at atmospheric pressure from 50 to 880°C? Use (a) direct integration of the C_p polynomial, (b) ICPH, and (c) MCPH. Report your answers in MJ.

Problem 4.9

A process stream is heated as a gas from 25 to 250°C at constant P. A quick estimate of the energy requirement is obtained from Eq. 4.3, with C_p taken as constant and equal to its value at 25 °C. Is the estimate of Q likely to be low or high? Why?

Problem 4.10

(a) Evaluate the latent heat of vaporization ΔH_v of n-pentane by Eq. 4.13. How does this result compare with the value listed in Table B.2?

(b) Handbook values for latent heats of vaporization at 25°C of four compounds are given in the table below. Calculate ΔH_v by Eq. 4.14, and compare the results with the values given in Table B.2.

Latent heats of vaporization at 25 °C in J/g

n-Pentane	366.3	Benzene	433.3
n-Hexane	366.1	Cyclohexane	392.5

Problem 4.12

Handbook values for the latent heat of vaporization in J/g are given in the table for several pure liquids at 0°C.

	ΔH at 0°C
Chloroform	270.9
Methanol	1,189.5
Tetrachloromethane	217.8

Calculate:

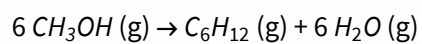
(a) The value of the latent heat at T_v by Eq. 4.14, given the value listed at 0°C.

(b) The value of the latent heat at T_v by Eq. 4.13.

By what percentage do these results differ from the value listed in Table B.2 of App. B?

Problem 4.20

Hydrocarbon fuels can be produced from methanol by reactions such as the following, which yields 1-hexene:



Compare the standard heat of combustion at 25°C of 6 CH_3OH (g) with the standard heat of combustion at 25°C of C_6H_{12} (g) for reaction products CO_2 (g) and H_2O (g).